Reproducibility Pipeline Project

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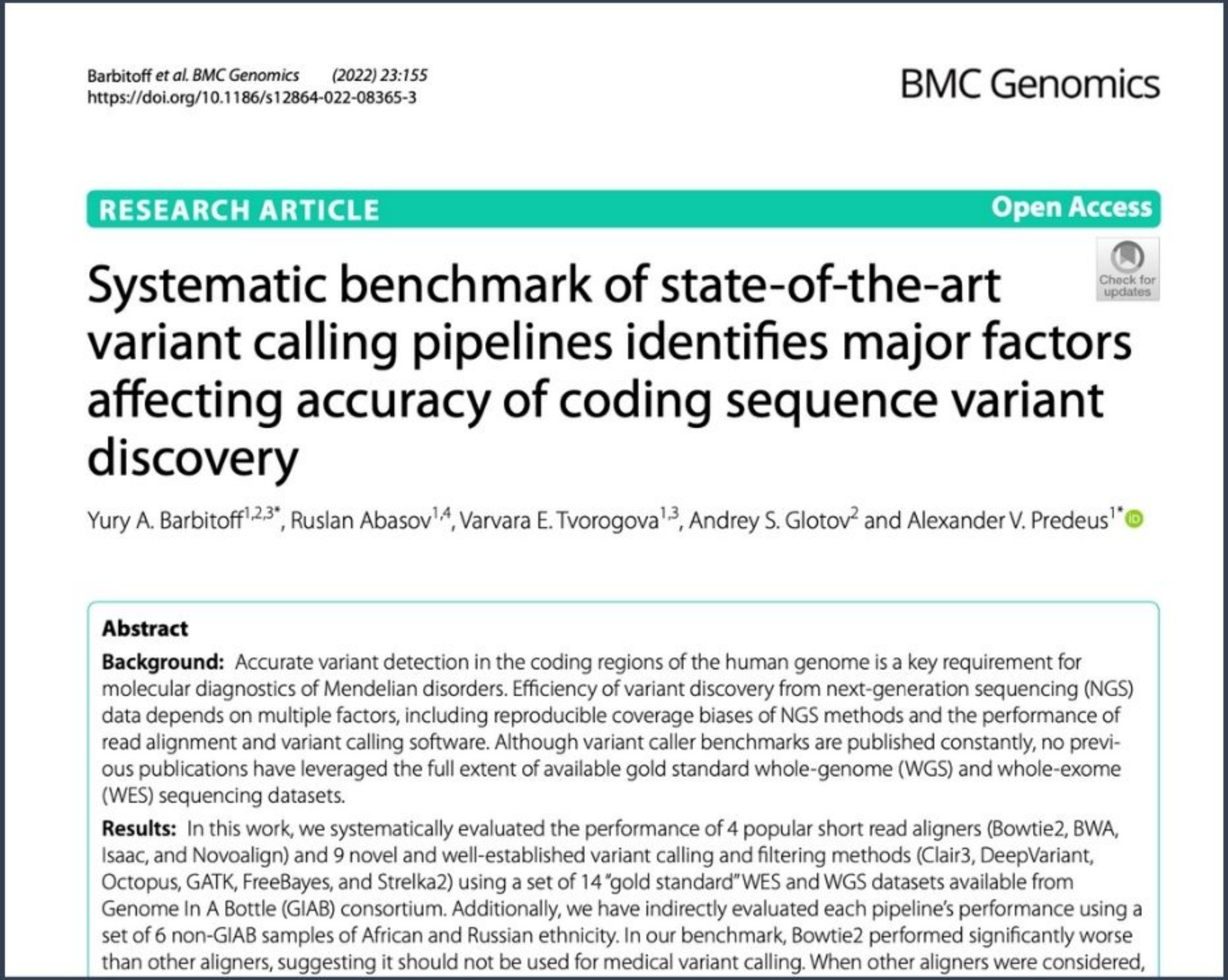
Replicating Barbitoff et al. (2022):



Barbitoff et al. BMC Genomics (2022) 23:155
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BMC Genomics

RESEARCH ARTICLE Open Access

Systematic benchmark of state-of-the-art variant calling pipelines identifies major factors affecting accuracy of coding sequence variant discovery

Yury A. Barbitoff[1,2,3*], Ruslan Abasov[1,4], Varvara E. Tvorogova[1,3], Andrey S. Glotov[2] and Alexander V. Predeus[1*]

Abstract

Background: Accurate variant detection in the coding regions of the human genome is a key requirement for molecular diagnostics of Mendelian disorders. Efficiency of variant discovery from next-generation sequencing (NGS) data depends on multiple factors, including reproducible coverage biases of NGS methods and the performance of read alignment and variant calling software. Although variant caller benchmarks are published constantly, no previous publications have leveraged the full extent of available gold standard whole-genome (WGS) and whole-exome (WES) sequencing datasets.

Results: In this work, we systematically evaluated the performance of 4 popular short read aligners (Bowtie2, BWA, Isaac, and Novoalign) and 9 novel and well-established variant calling and filtering methods (Clair3, DeepVariant, Octopus, GATK, FreeBayes, and Strelka2) using a set of 14 "gold standard" WES and WGS datasets available from Genome In A Bottle (GIAB) consortium. Additionally, we have indirectly evaluated each pipeline's performance using a set of 6 non-GIAB samples of African and Russian ethnicity. In our benchmark, Bowtie2 performed significantly worse than other aligners, suggesting it should not be used for medical variant calling. When other aligners were considered,

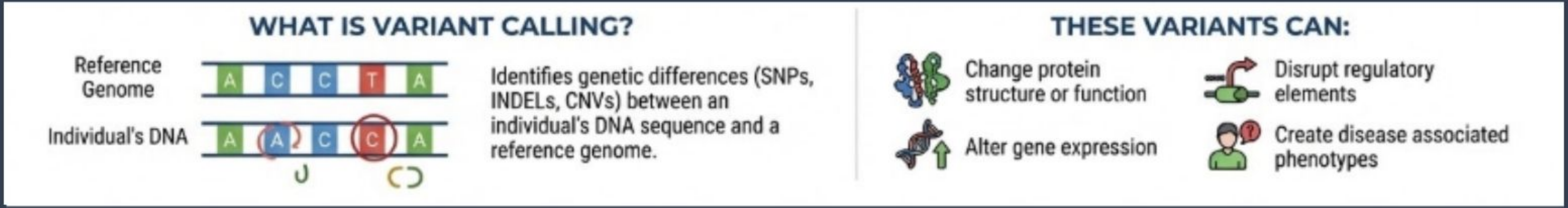
Key Themes

Accuracy of variant calling impacts diagnosis, research, and precision medicine

Comparing two major pipelines: DeepVariant vs GATK

Importance of reproducibility in computational genomics

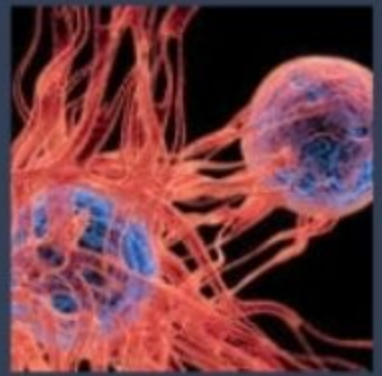
Key Applications of Variant Calling



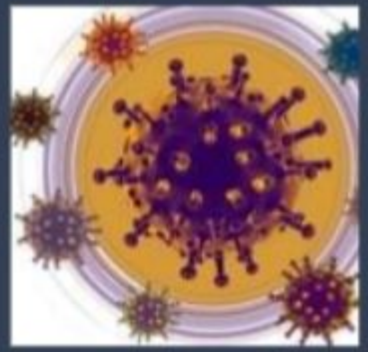
Clinical Research & Trials

Population Health

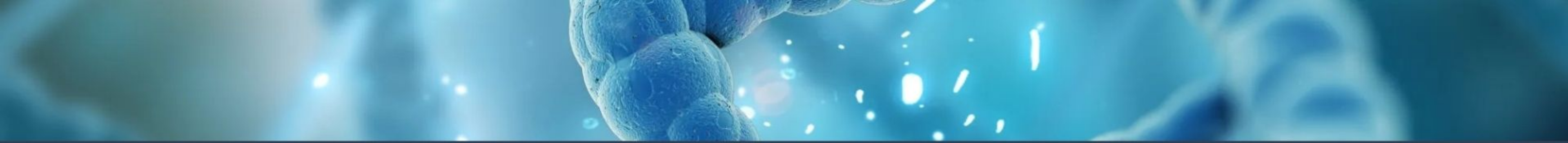
Pharmacogenomics



Cancer Precision Medicine

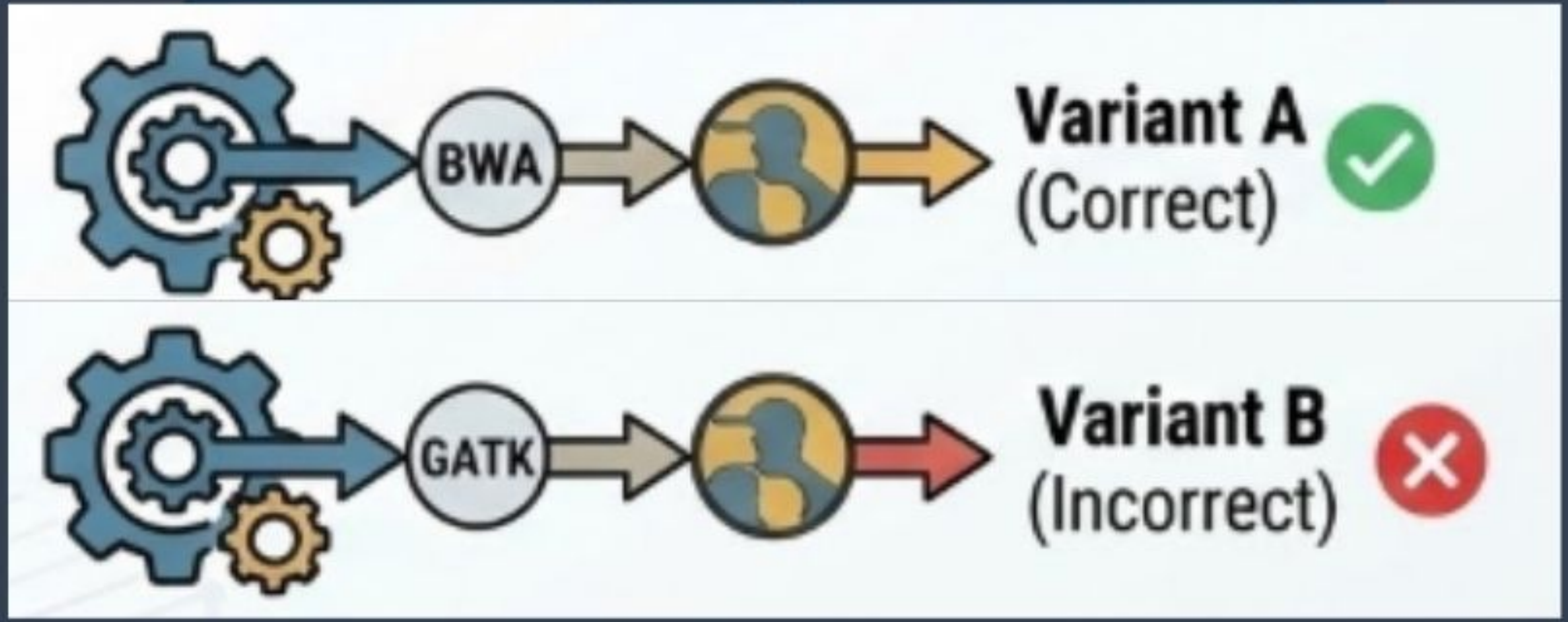


Genetic Diagnosis



Background & Statemtent of Need

The Challenge: Pipeline Variability



The Need: Reproducibility & Benchmarking

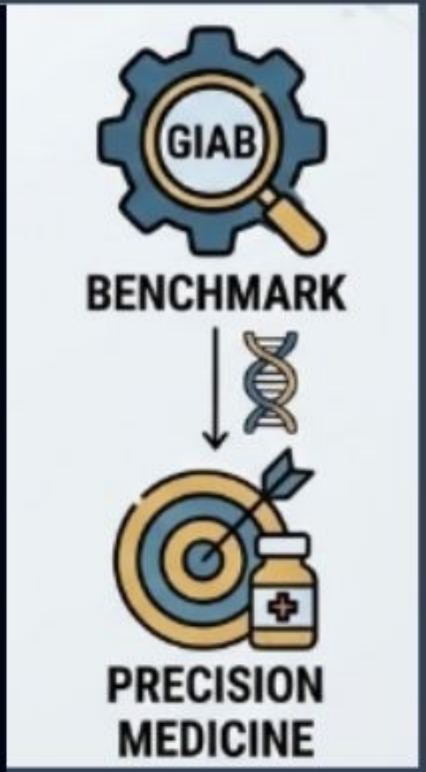
The Knowledge Gap

Prior to this systemic comparison, the field lacked a large-scale analysis of variant callers across multiple samples.

The "NA12878" Limitation

Earlier studies focused heavily on a single sample (NA12878). This narrow focus fails to account for:

- Behavior across different genomic regions
- Varied sequencing conditions
- Systemic biases in specific pipelines



Clinical Implications

⚠ CRITICAL IMPLICATIONS

When Pipelines Fail:
The Diagnostic Conflict

Pipeline choice isn't just technical—it's clinical. Conflicting results can alter the course of cancer research and personalized medicine.

Case Study: (Skitchenko et al., 2024) Missed Diagnosis

A clinical pipeline **failed to detect a pathogenic mutation**.

HaplotypeCaller (GATK) → DeepVariant

Patient told "Negative" — Later identified by different tool

PATHOGENIC_VARIANT_NOT_FOUND

ERR_PIPELINE_SENSITIVITY

Pipeline Discordance

According to Hwang et al. (2015)*, agreement between variant calling tools is far from perfect. This divergence necessitates a deeper understanding of which factors (read depth, aligner algorithm, variant caller logic) drive the differences.

** Note: While referenced here, newer studies continue to refine these discordance rates.*

DISAGREEMENT RATE

>20%

Frequency tools disagree on high-quality sequencing data

Original Researcher's Hypothesis I

Caller vs. Aligner

The researcher hypothesized that the choice of **Variant Caller** has a significantly larger effect on accuracy than the choice of Aligner.

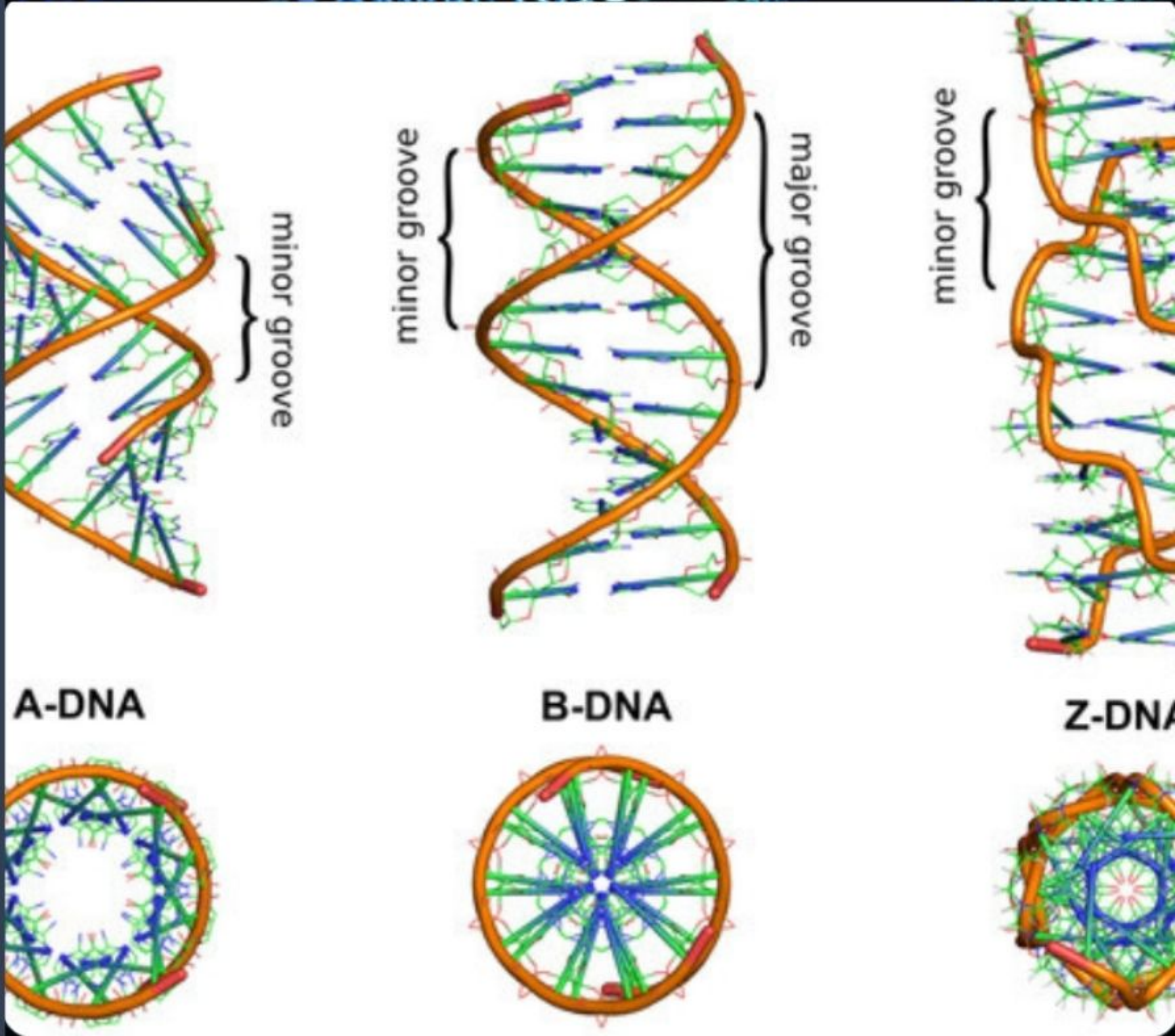
ML vs. Traditional

Modern machine-learning tools (e.g., DeepVariant) were expected to outperform traditional statistical methods like GATK.

Genomic Context

Accuracy is not uniform; it is strictly dependent on genomic context, such as GC content and exon boundaries.

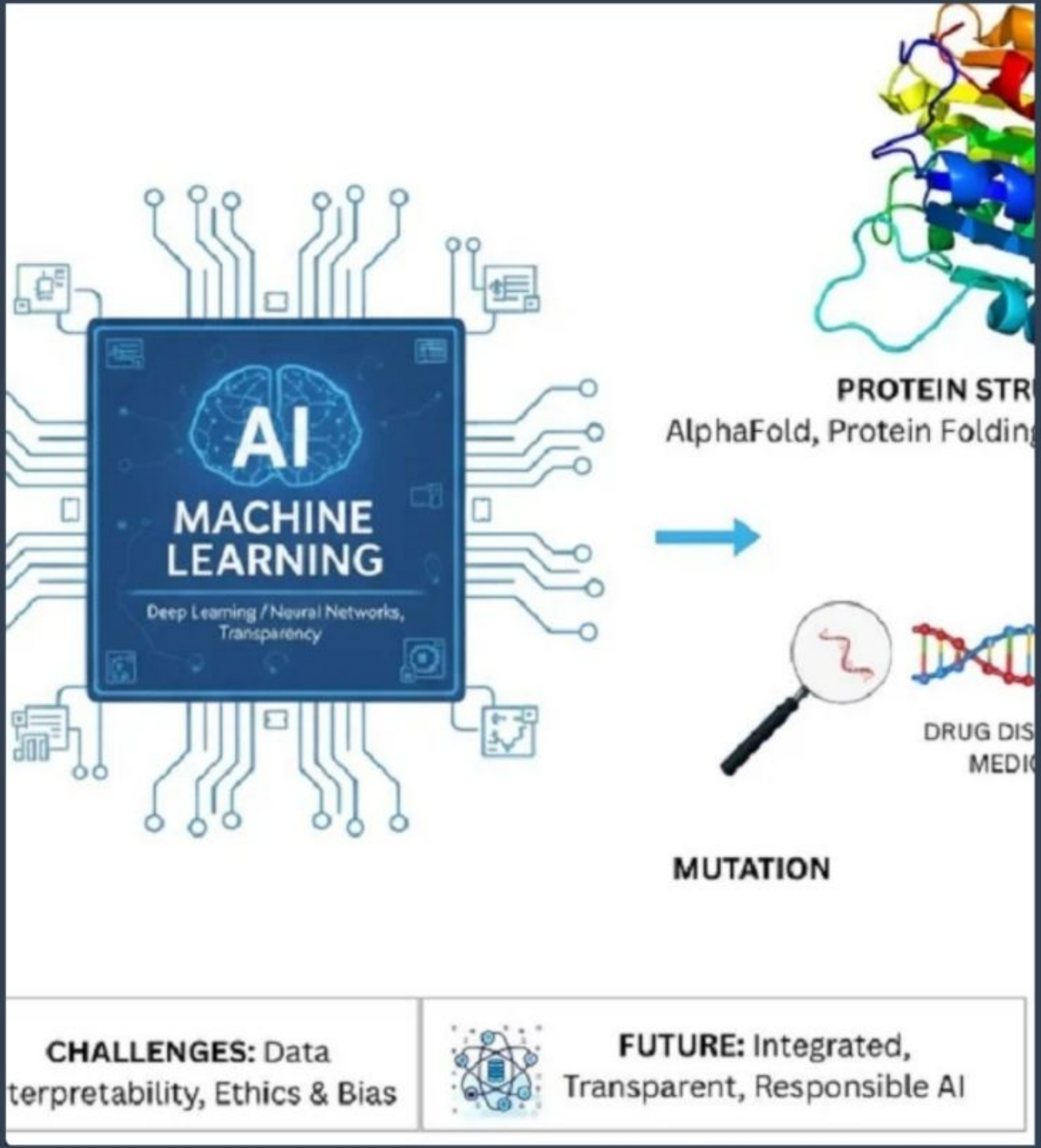
Genomic Context
Where do pipelines fail?



Accuracy fluctuates

- Physical properties of the DNA region being sequenced;
    - High/low GC Content are notoriously difficult to sequence & align
    - Exon Boundaries are prone to alignment artifacts

Original Researcher's Hypothesis II



The study specifically aimed to validate the shift towards AI-driven analysis.

- Traditional (GATK): Relies on hand-crafted statistical models and filters.
- Modern (DeepVariant): Uses deep neural networks to "see" variants in read data, hypothesizing superior sensitivity and specificity.

Evaluating Variant Calling Pipelines

Replication of Barbitoff et al. (2022)

WHAT THE ORIGINAL PAPER DID

Benchmarked DeepVariant vs GATK across real WES + WGS GIAB datasets

Tested 45 pipeline combinations (multiple aligners, filters, preprocessing)

Found DeepVariant has higher sensitivity, while GATK is more conservative

WHAT WE DID

Reproduced key findings using synthetic normal + tumor datasets

Injected known somatic mutations for controlled benchmarking

Compared DeepVariant vs GATK using the same modern tools

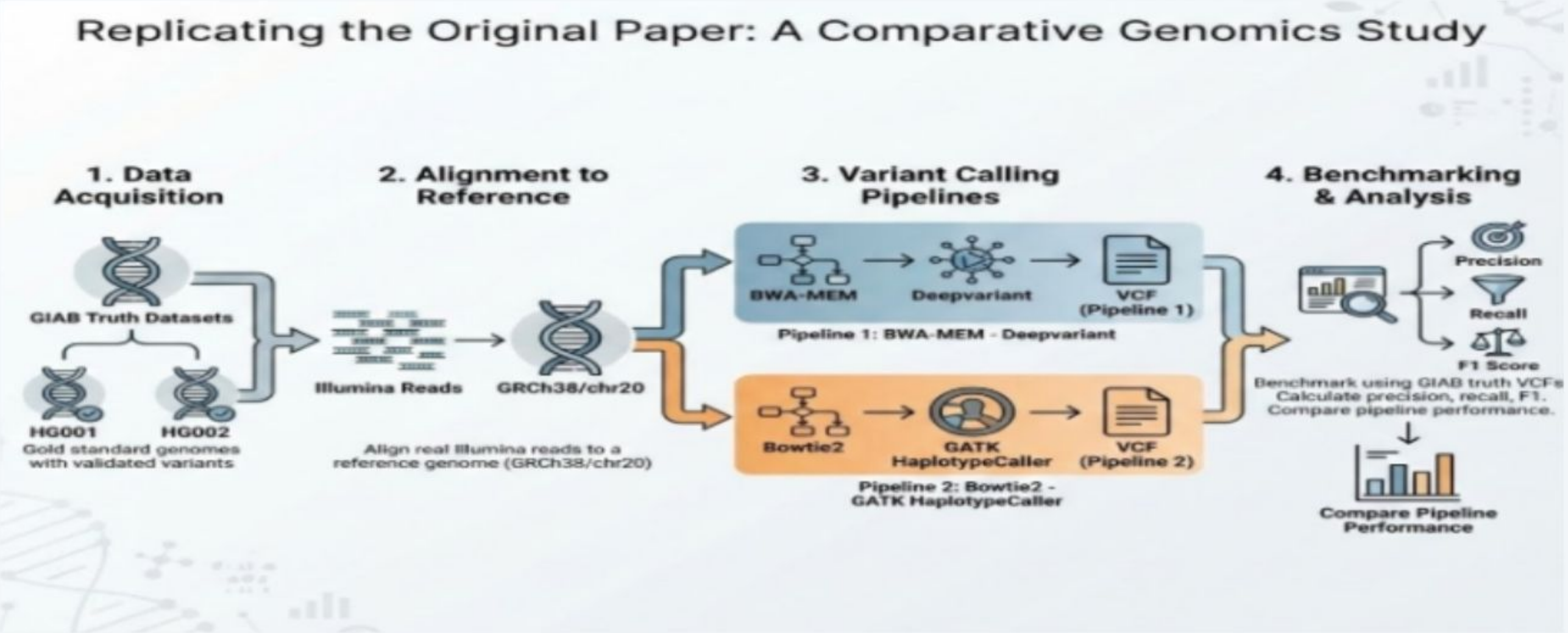
Evaluate accuracy, sensitivity, and concordance

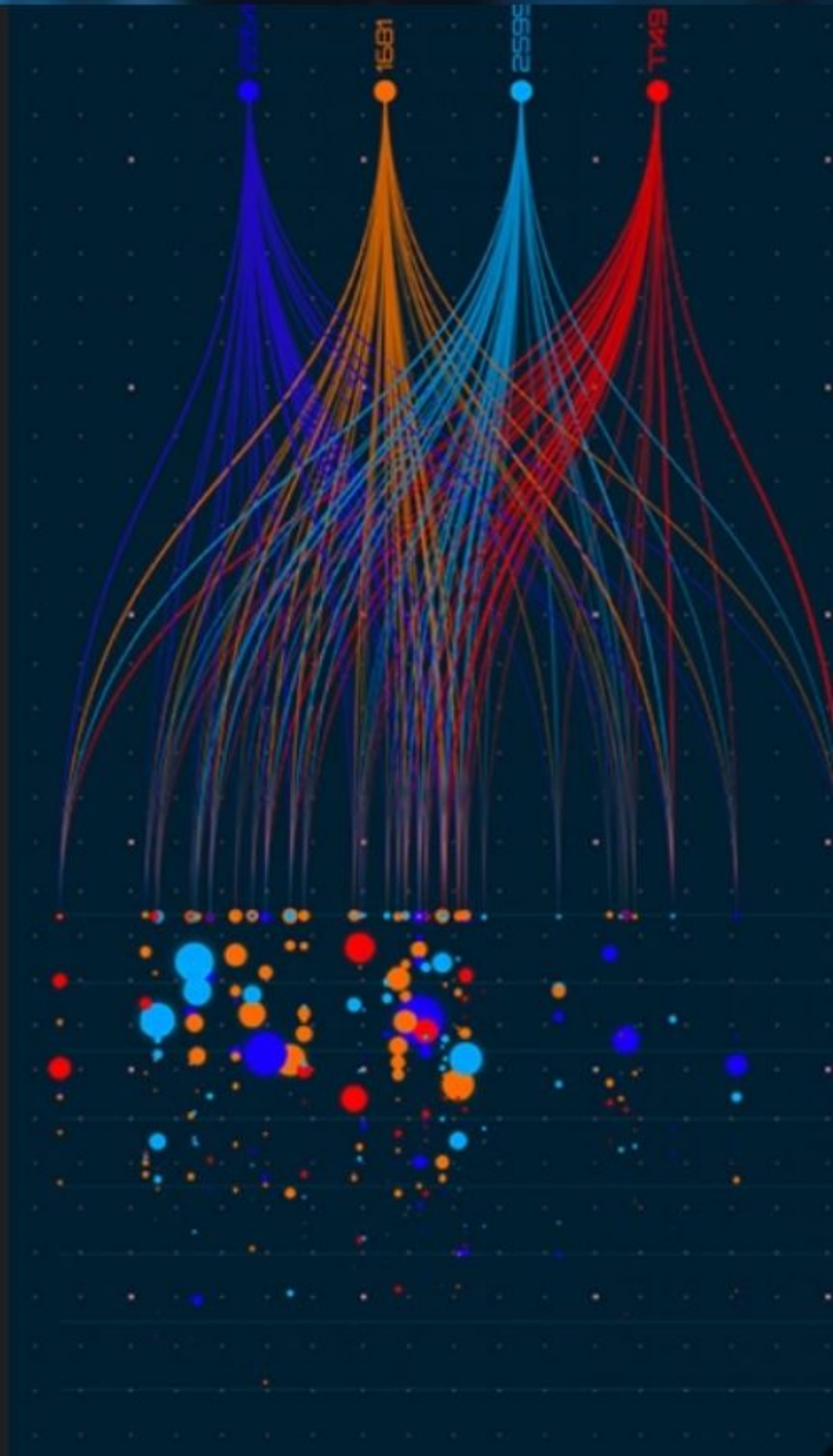
Assessed whether trends from the original study still hold

OVERALL GOAL

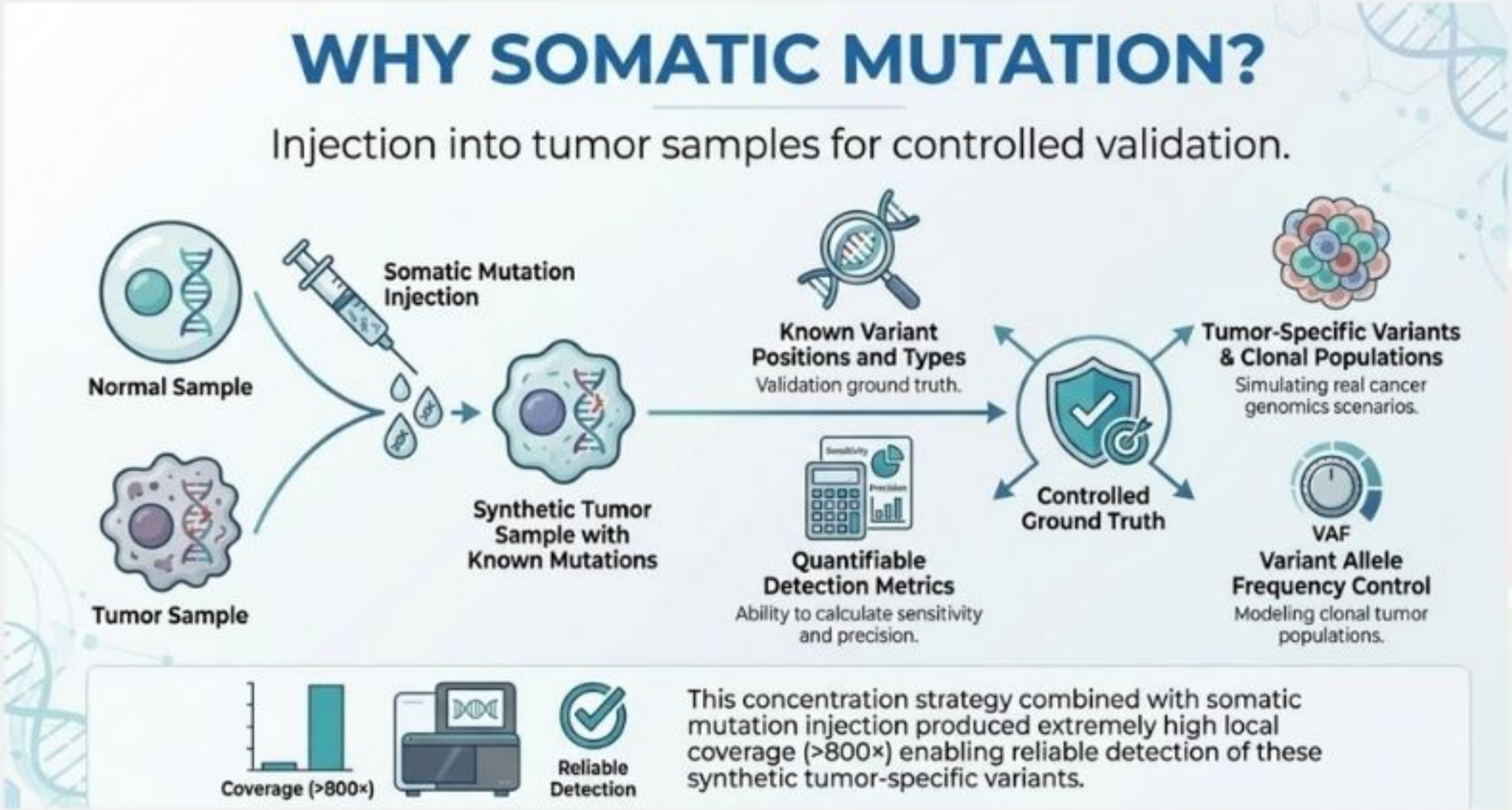
Do DeepVariant and GATK perform the same way in our environment as they did in the original study?

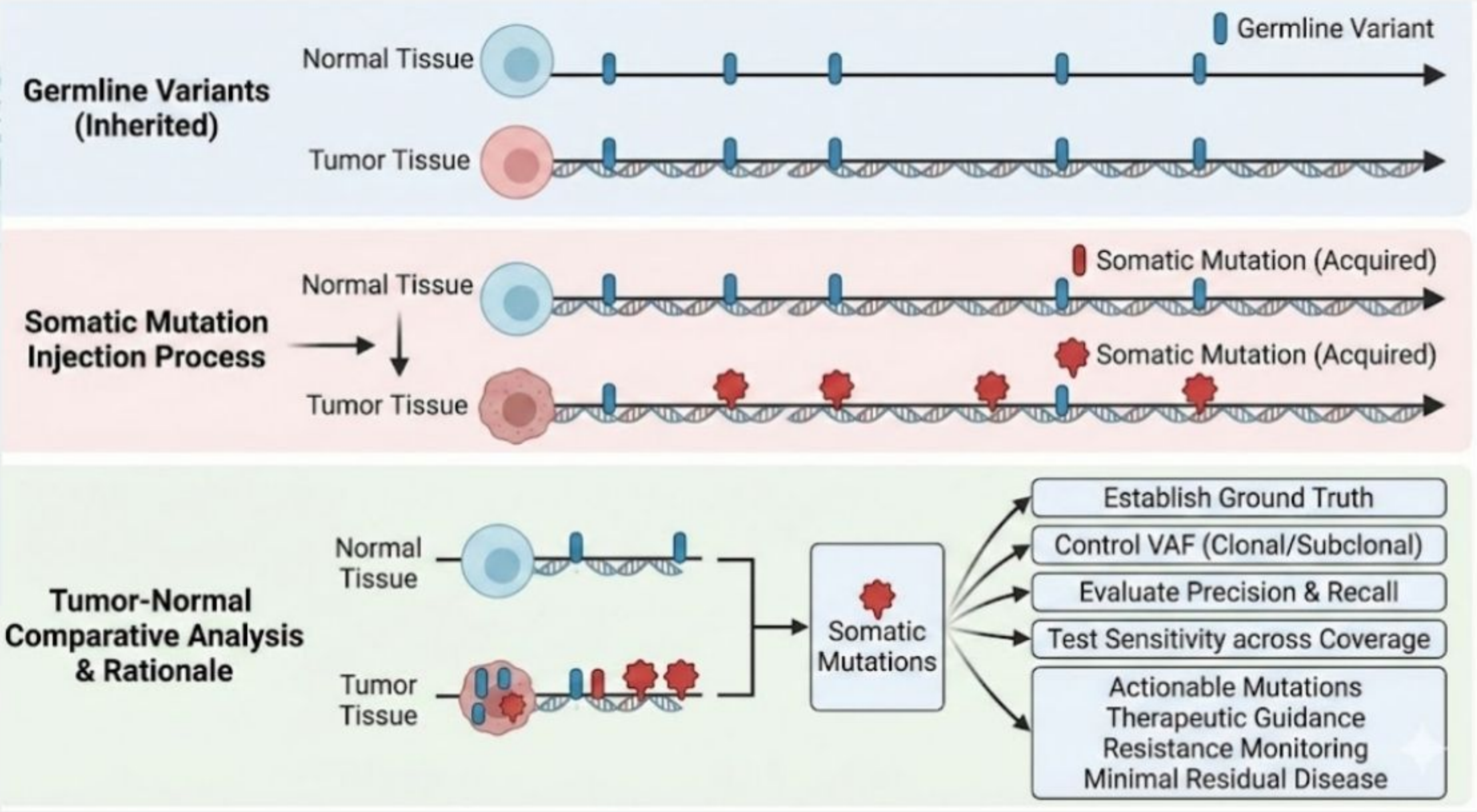
Setting our environment

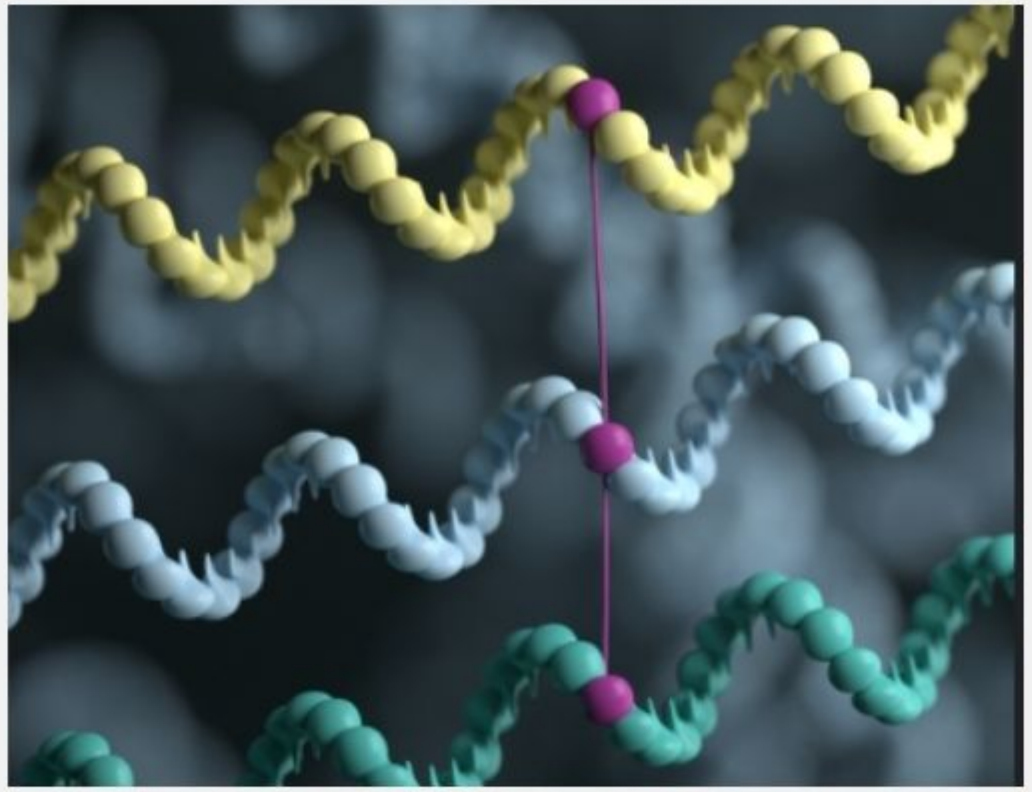




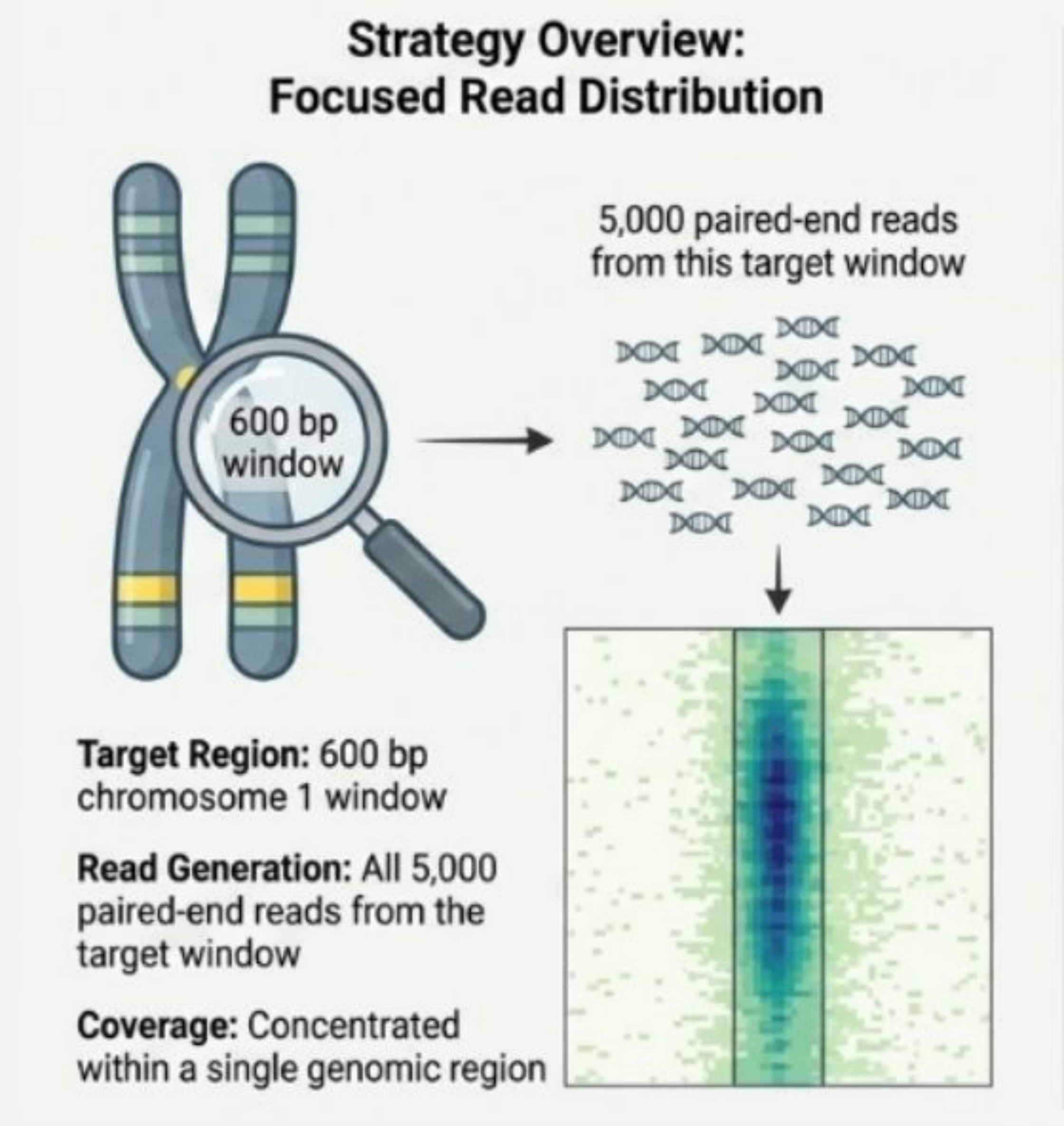
Synthetic Read Generation







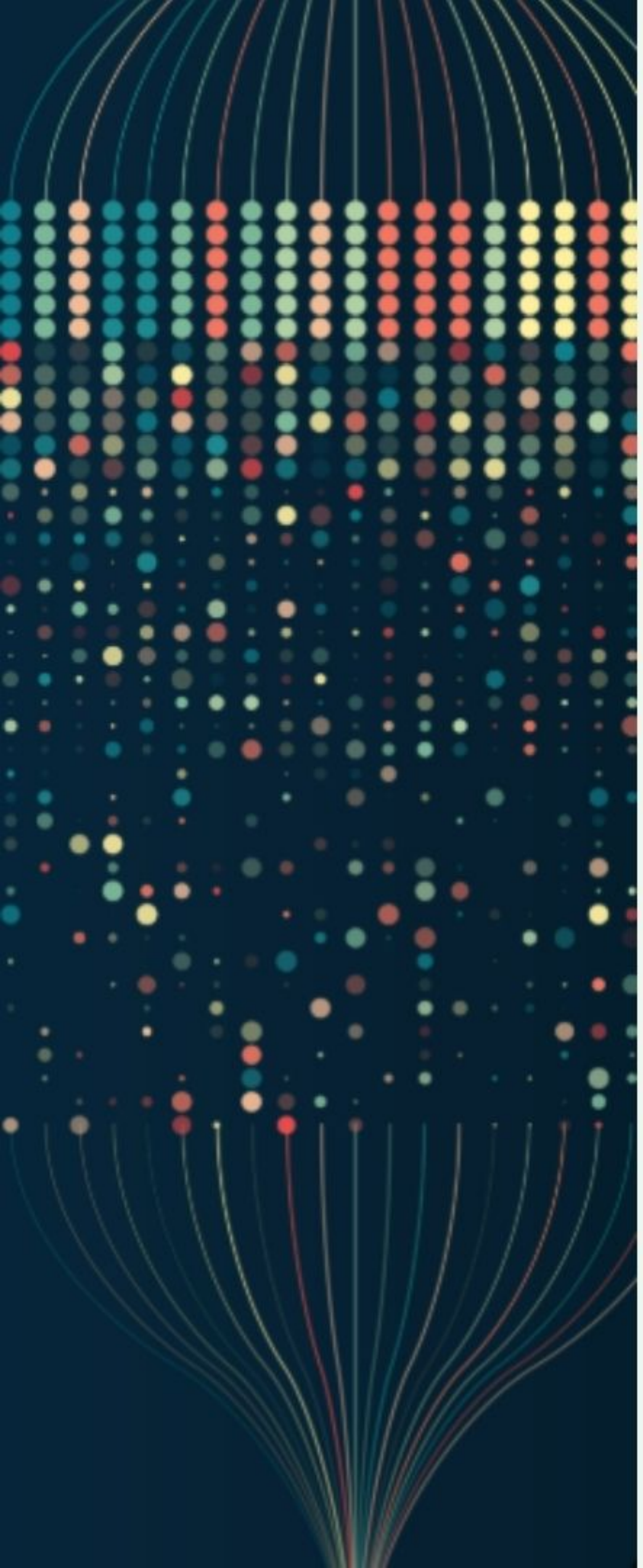
Had we simply generated reads from the reference genome without mutation injection, both normal and tumor samples would have been identical, providing no distinguishable variants to detect and no basis for benchmarking variant caller performance.



DATASET 1

The first dataset implemented a focused read distribution approach targeting the 600 bp chromosome 1 window. This strategy generated four independent FASTQ files representing paired tumor-normal samples with biological replicates:

- normal1_R1/R2.fastq
- tumor1_R1/R2.fastq
- normal2_R1/R2.fastq
- tumor2_R1/R2.fastq

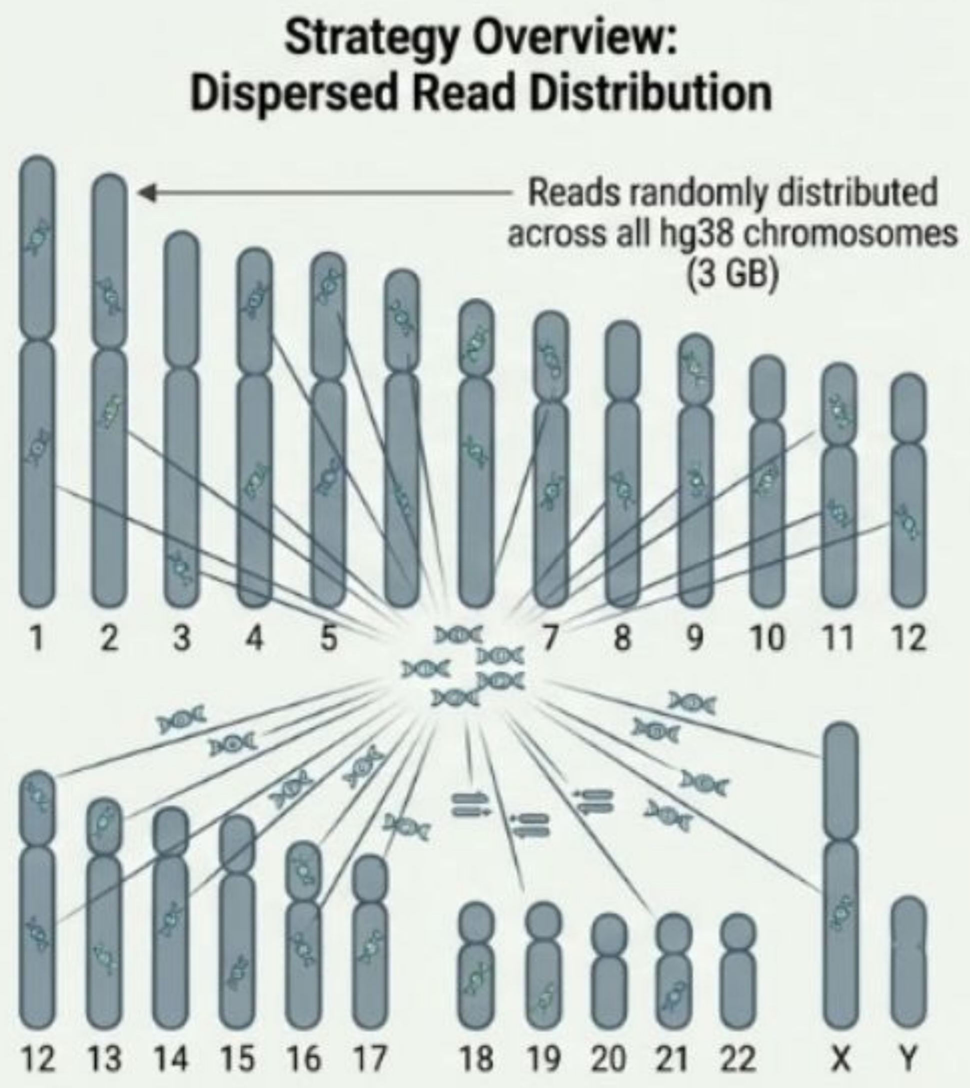


DATASET 2 & 3

Dispersed coverage distribution, we generted two additional datasets (Datasets 2 and 3) using an alternative strategy. Rather than genomic fragments from across the entire hg38 reference genome (3 GB).

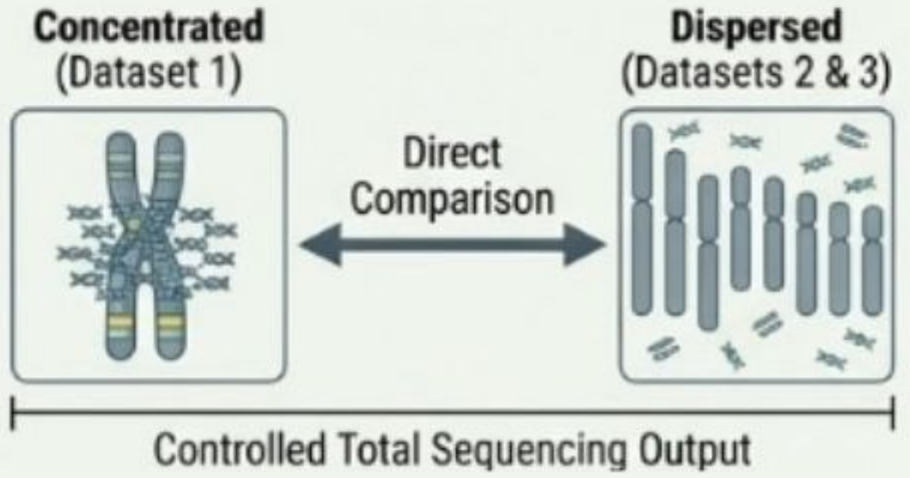
Key Features:

- Some total read count: 5,000 paired-end reads per sample
- Reads randomly distributed across all chromosomes
- Same mutation injection protocol for tumor samples (2% mutation rate)
- Identical read characteristics (150 bp length, 350 bp insert size)
- Same quality score distributions and FASTQ formatting

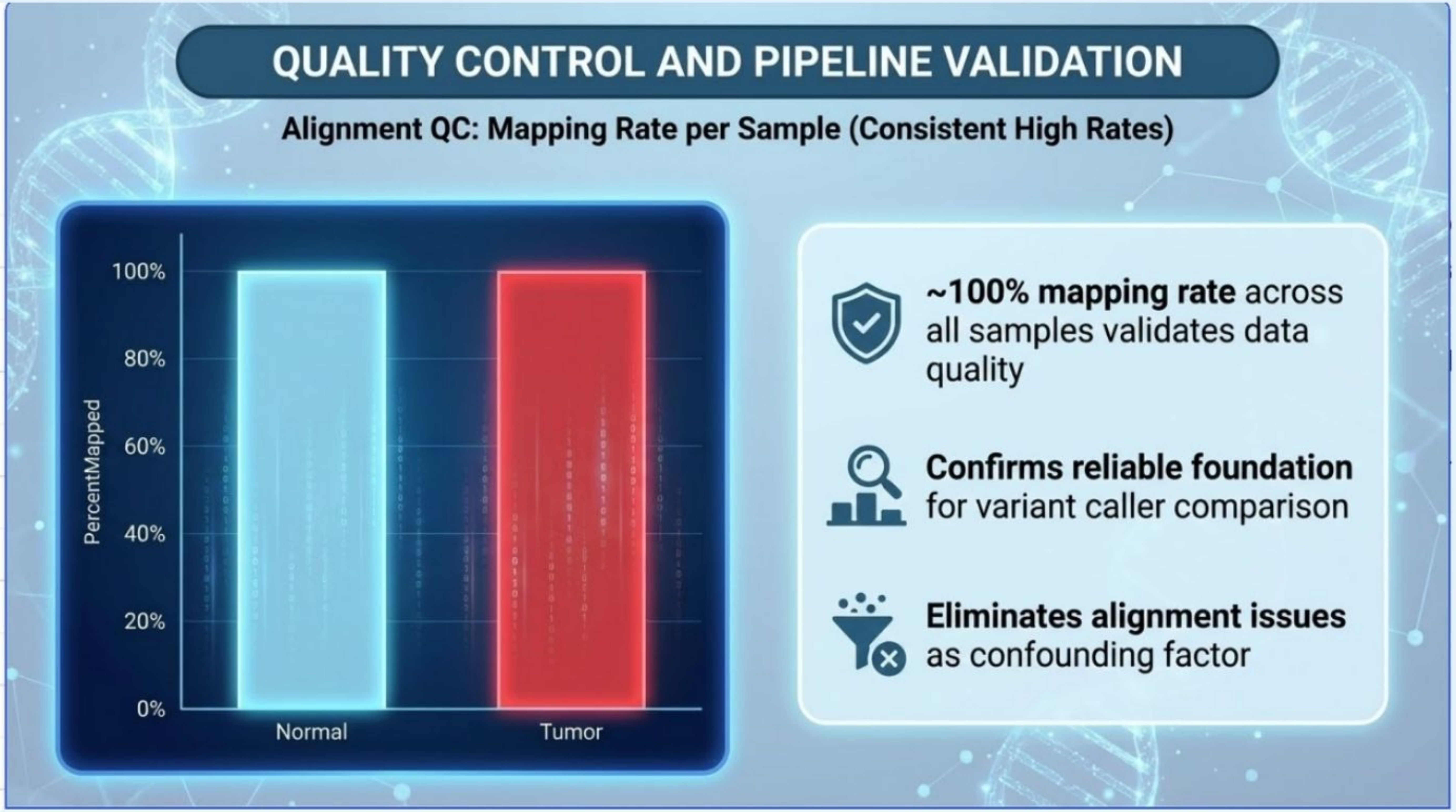


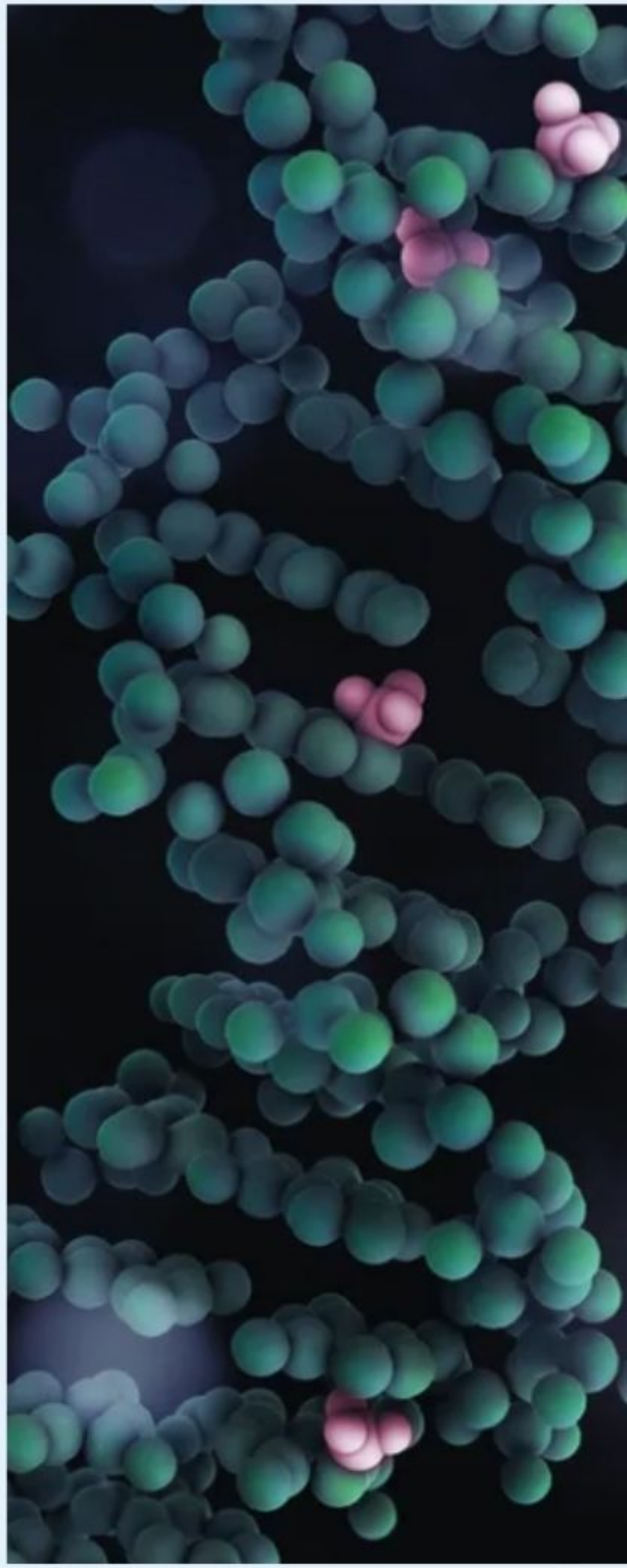
Experimental Design Purpose

This design enables a direct comparison between concentrated (Dataset 1) and dispersed (Datasets 2 & 3) read distribution strategies while controlling for total sequencing output.

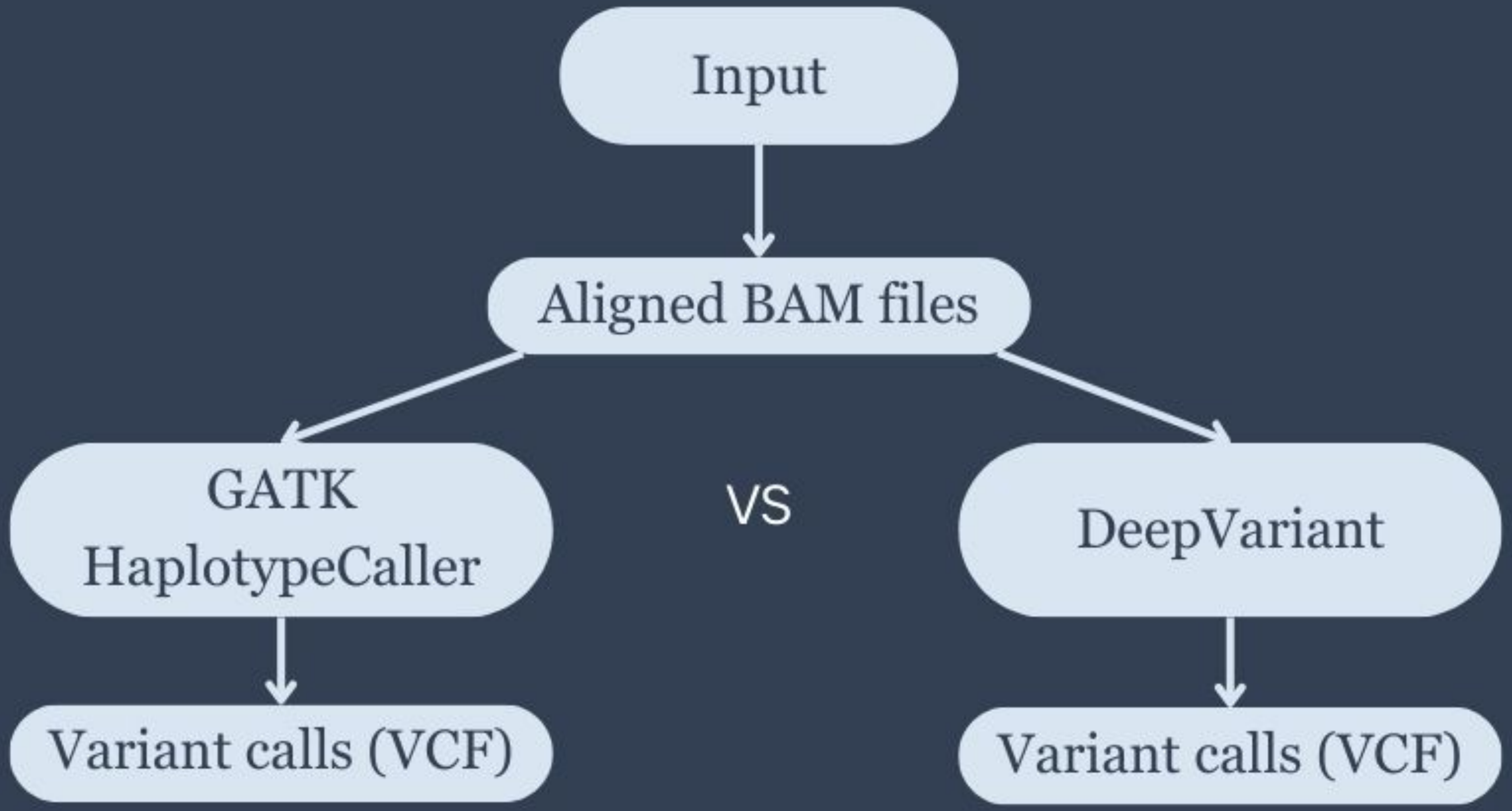


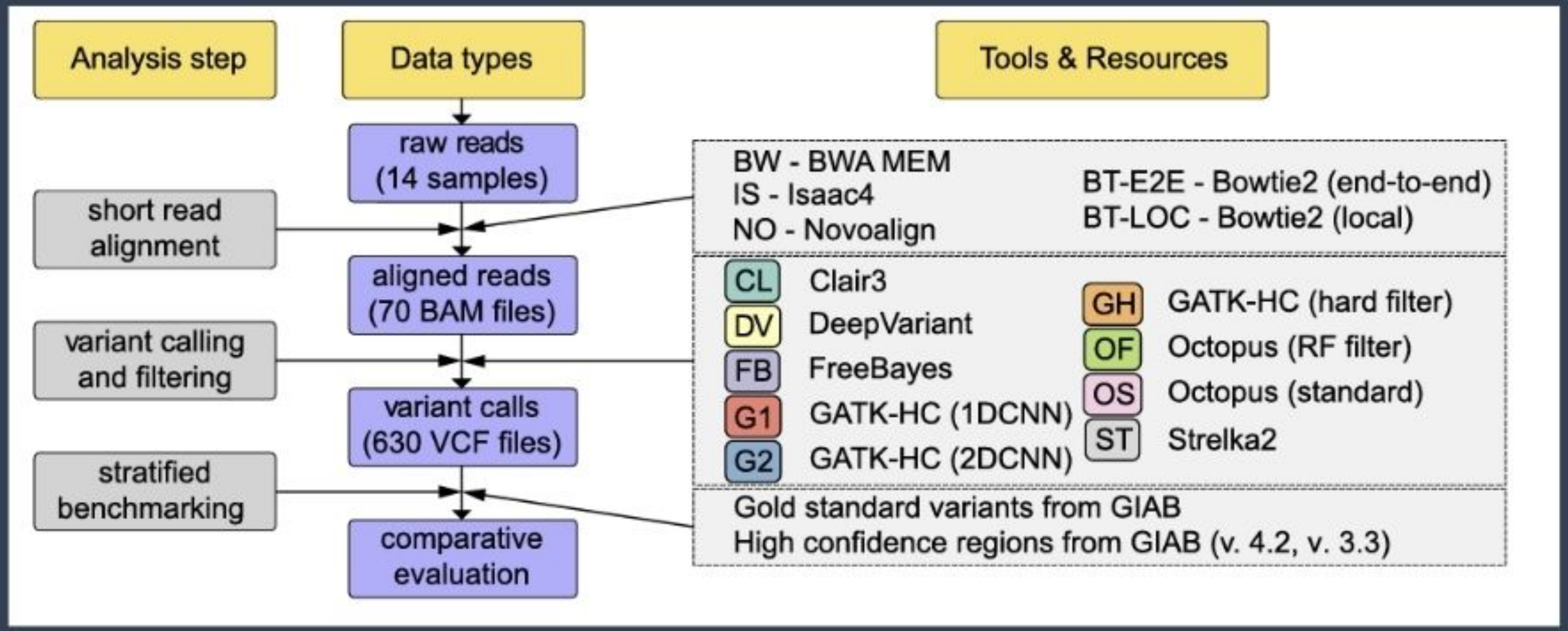
QUALITY CONTROL AND PIPELINE VALIDATION





Variant Calling Pipeline





Goal: Reproduce the paper's finding; DeepVariant detects more variants than GATK.

Variant Calling - HPC

Barbitoff et al. (2022) Study Workflow	Our Environment
• Docker Containers • GATK 4.2.3.0 & DeepVariant 1.2.0	• No Docker on HPC • Need conversion

Solution

- singularity pull docker://broadinstitute/gatk:4.2.3.0
- singularity pull docker://google/deepvariant:1.2.0

Result

- Same exact software versions
- Reproducible analysis

Issues faced:

- Reference mismatch → Updated from GRCh37 to hg38
- Missing read groups → Added RG tags to BAM files
- Container can't find files → Fixed bind mounting: --bind /scratch:/scratch

Variant Calling Results

Sample	GATK	DeepVariant	DV Advantage
Synthetic_NORMAL	172	197	+25 (+14.5%)
synthetic_TUMOR	377	408	+31(+8.2%)

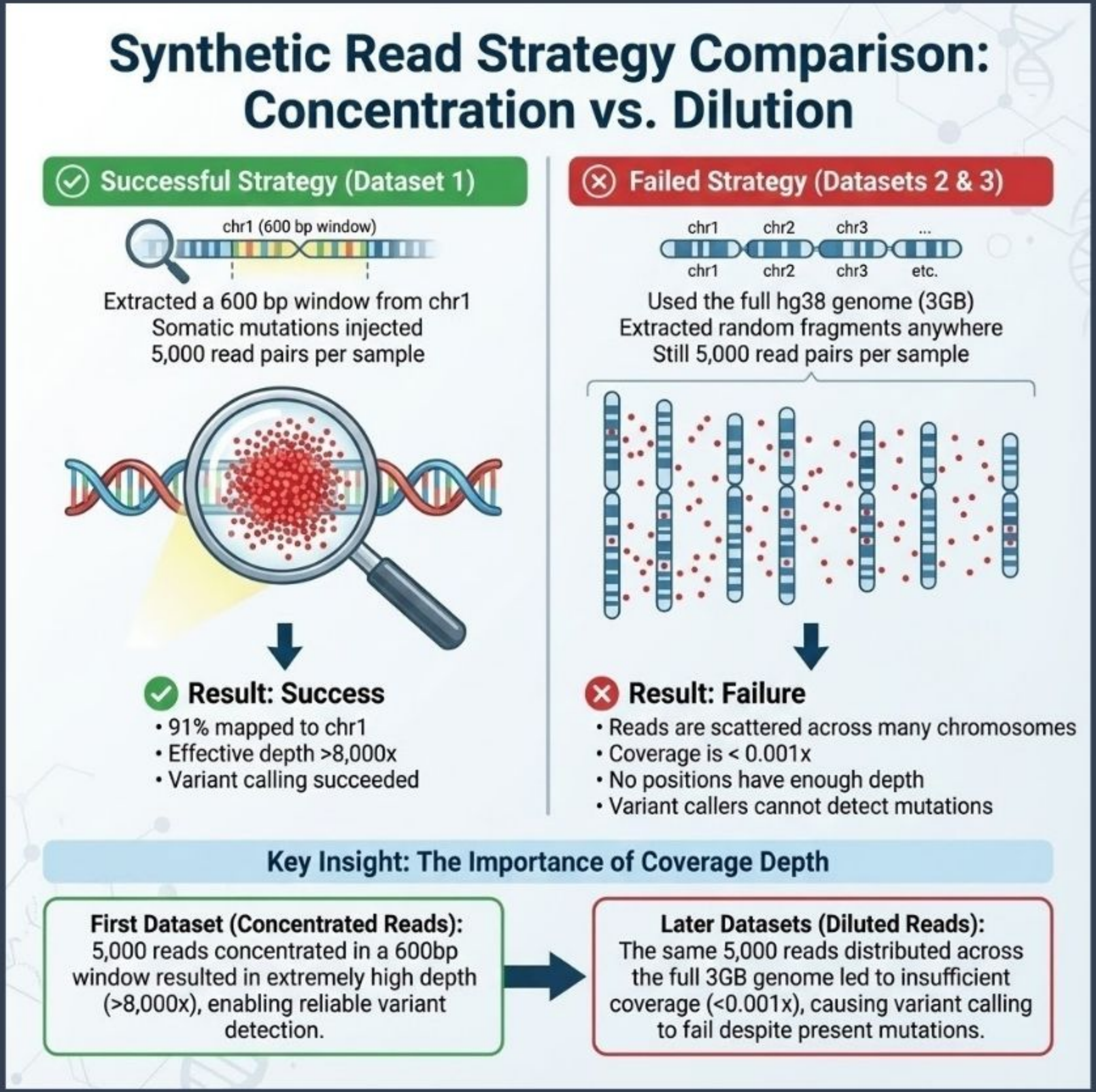
DeepVariant detects **14.5%** more variants on Normal

DeepVariant detects **8.2%** more variants on Tumor

This is consistent with published results, where DeepVariant tends to recover additional SNPs with better sensitivity, particularly in low-complexity and noisy regions, while GATK is more conservative

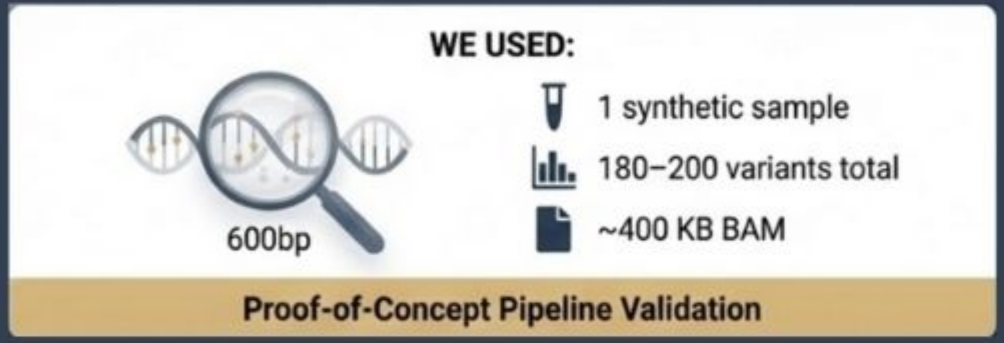
Datasets Coverage

- Read concentration matters as much as total read count.
- Both have 5,000 reads but dataset 1 succeeded but others didn't.
- Local coverage depth is crucial for variant calling not just high total sequencing output.



Variant Count

Barbitoff et al. (2022) Study Workflow	WGS 30x (Whole Genome Sequencing)
Our Environment	~ 800x on 600bp



Smaller genome region

Small mutations

Localized coverage

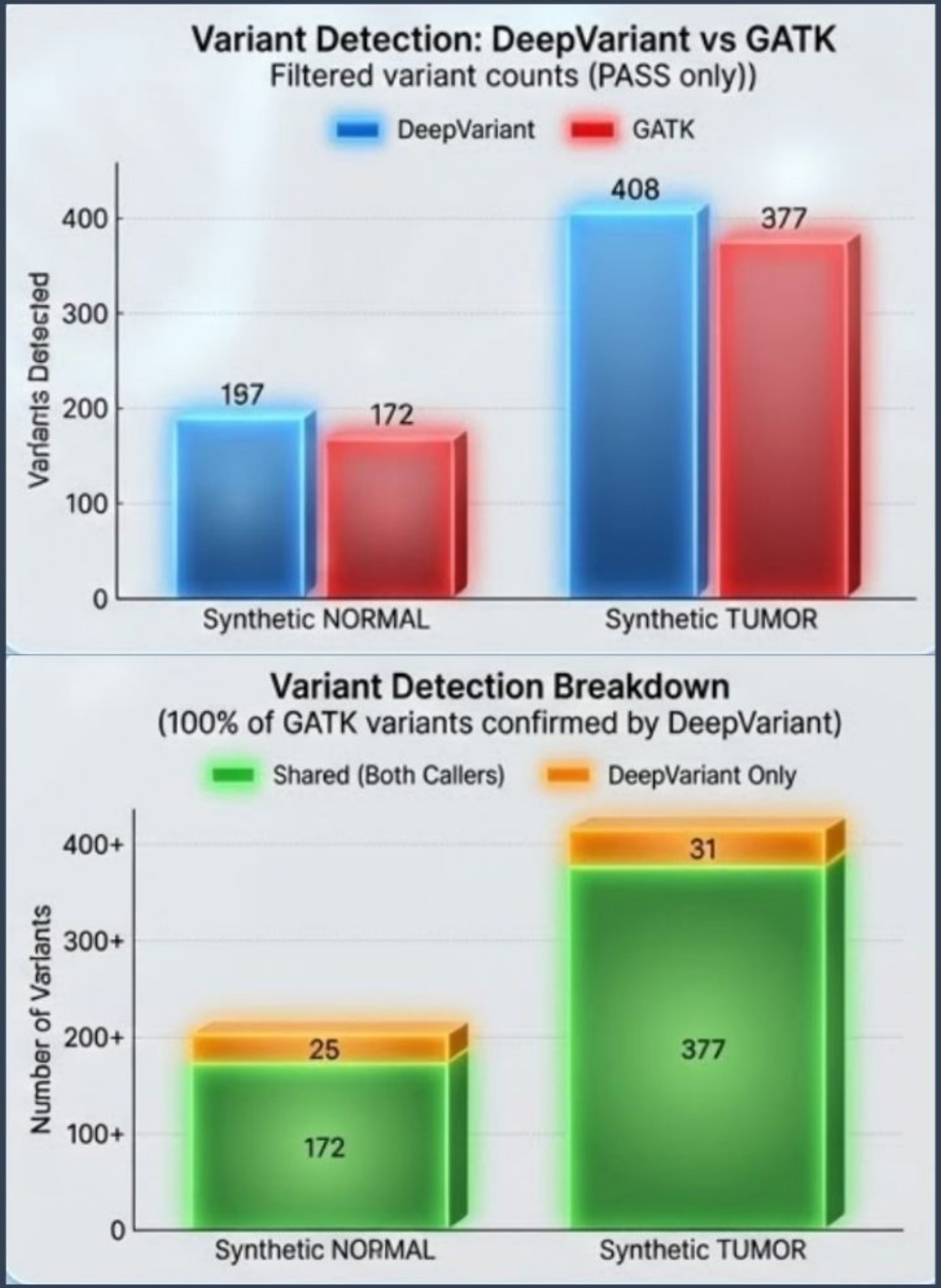
Pipeline validation

Study Comparison

Category	Barbinoff et al. (2022) Study	Our Work
Data Scale	Full GIAB WGS datasets	Synthetic reads targeting 600bp region
BAM size	40–80GB per sample	400–800KB per sample
Number of samples	14	1 synthetic sample (Normal + Tumor)
Variants/sample	20,000–25,000	180–200
Pipelines	45 pipelines (9 callers × 4 aligners × filters)	2 pipelines (GATK vs DeepVariant)
Aligners tested	BWA, Bowtie2, Isaac, Novoalign	BWA (default)
GATK pipeline	BQSR + HaplotypeCaller + CNN/Hard filters	HaplotypeCaller + hard filtering only
DeepVariant models	GIAB WGS/WES models + gVCF	Same DeepVariant model, VCF only

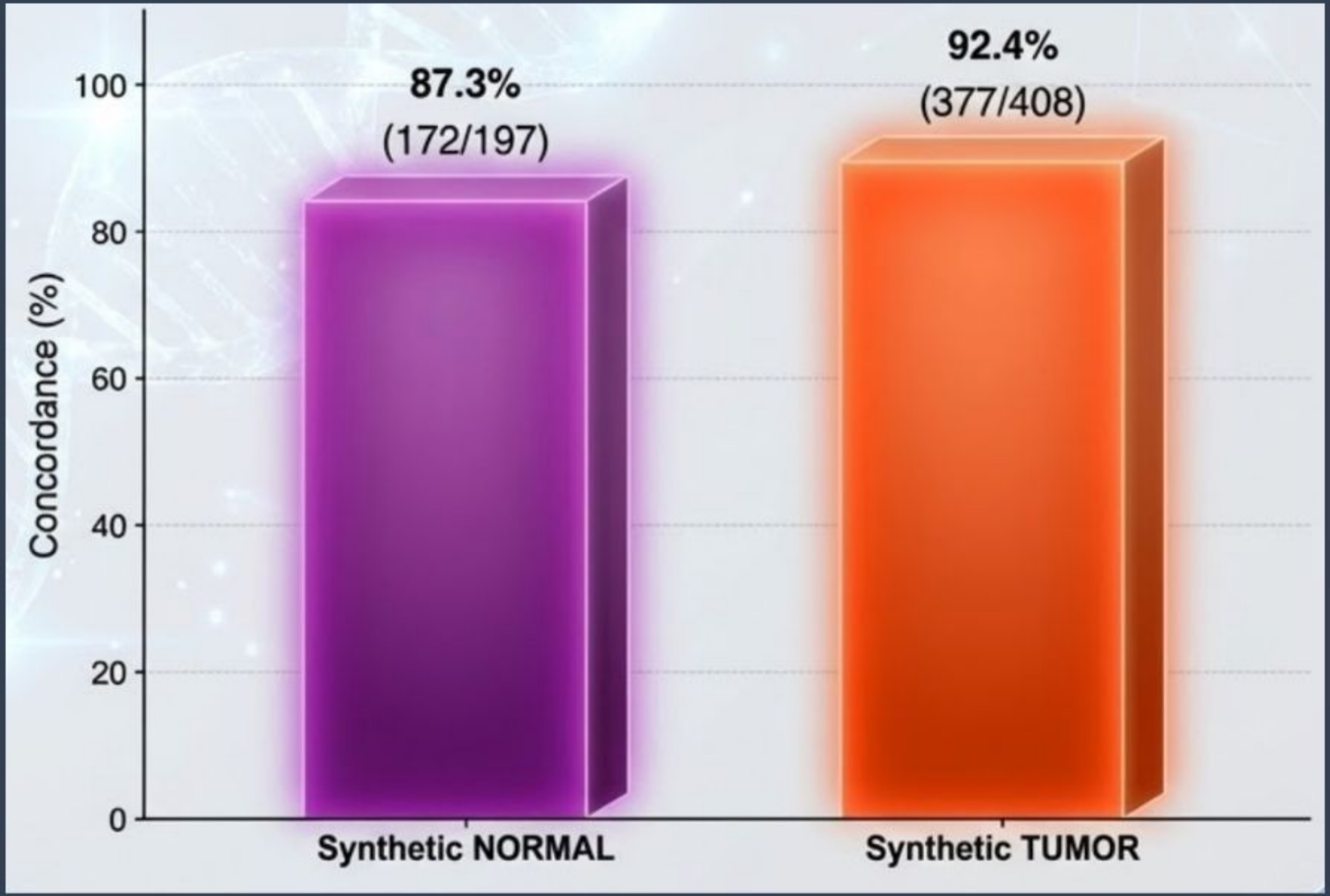
Variant Detection Comparison: DeepVariant vs GATK

- GATK-only variants = 0 (GATK ⊆ DeepVariant)
- DeepVariant finds everything GATK finds + 6-14% more
- Confirms DeepVariant's higher sensitivity from original paper
- Machine learning captures patterns traditional statistics miss
- GATK's conservative approach prioritizes precision over recall



Pipeline Agreement Analysis: Concordance Percentages

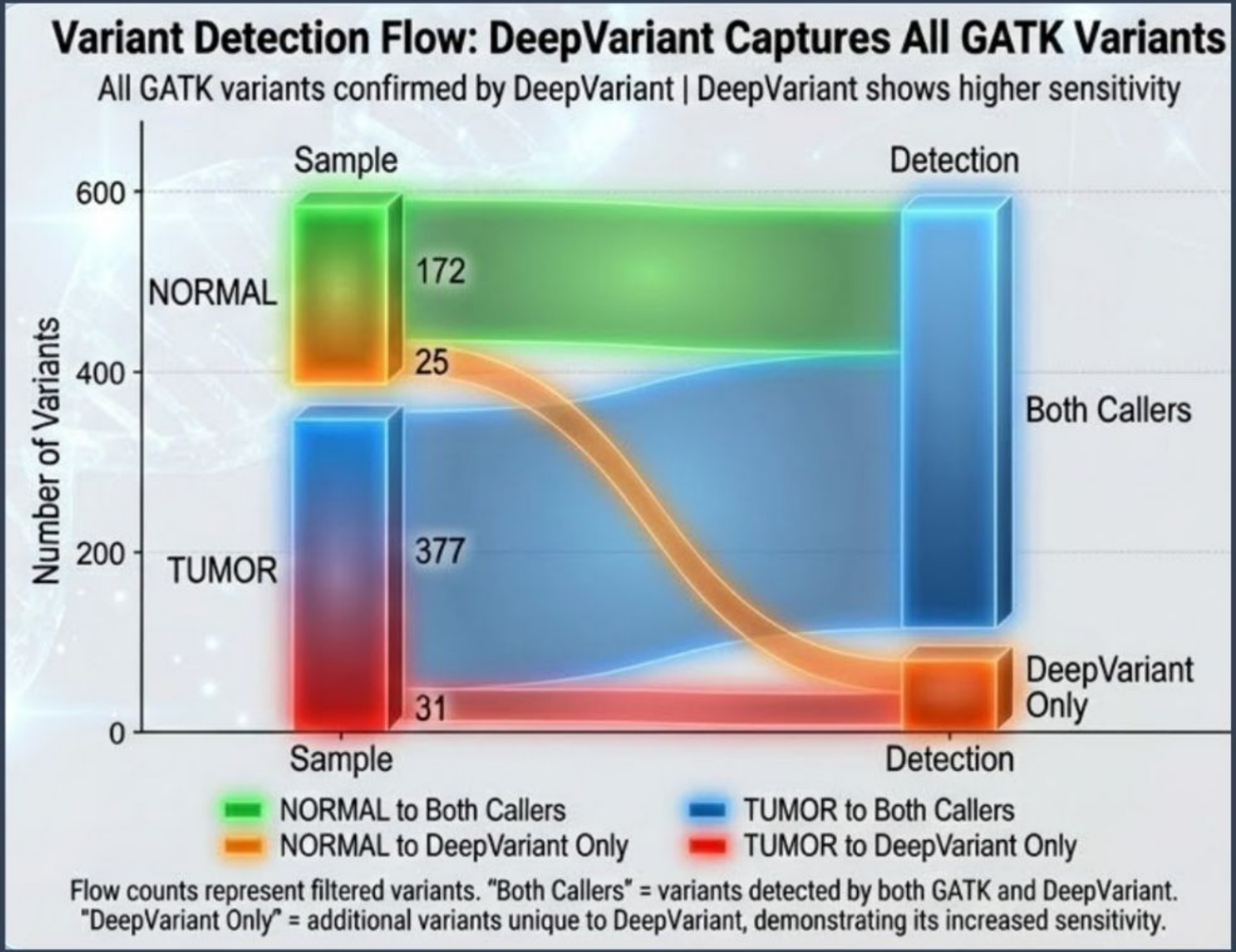
Strong Concordance Validates Both Tools



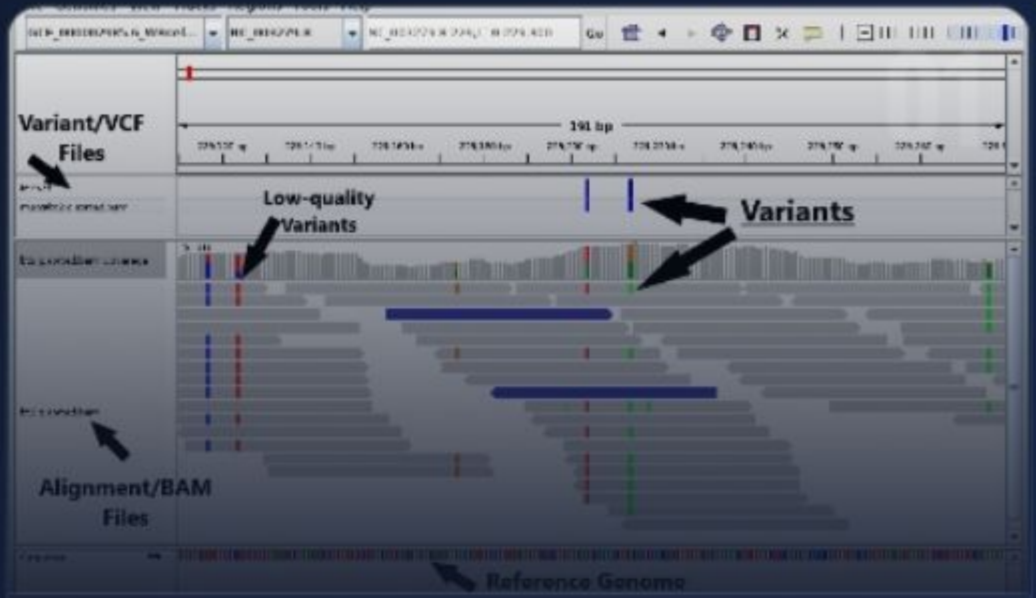
- Pipelines agree on core variants; differences are edge cases
- All GATK calls confirmed by DeepVariant (100% overlap)
- High agreement; both tools are detecting true biology
- ~8-13% discordant calls represent ambiguous/low-coverage region
- Suggests **DeepVariant for discovery, GATK for validation**

Complete Variant Calling Performance

- 100% of GATK variants confirmed by DeepVariant
- Visual flow shows no GATK-exclusive detection pathway
- Validates Barbitoff et al. (2022): DeepVariant > GATK
- Reproduced on synthetic data with identical software versions
- Complete subset relationship (GATK ⊆ DV) provides strong evidence



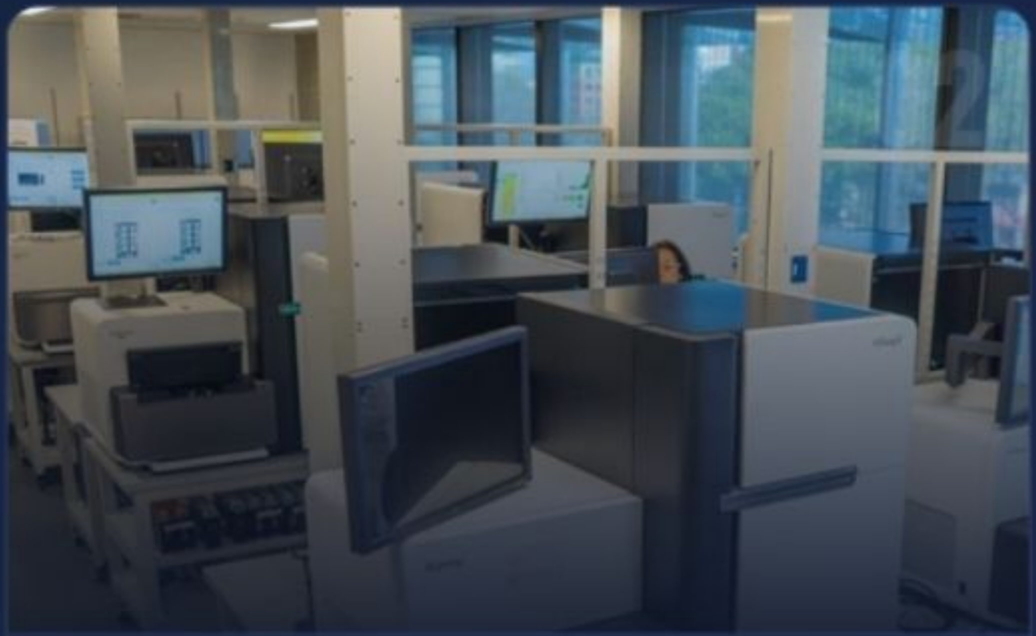
Variant Caller Choice Matters



HIGH SENSITIVITY

DeepVariant outperforms GATK, detecting significantly more valid variants.

+6-14% detection rate



APPLICATION

Research: Prioritize DeepVariant for discovery.
Clinical: Use an ensemble approach for validation.

FUTURE VALIDATION

Next steps include full WGS dataset testing, computational cost analysis, and structural variant extension.

Takeaway: DeepVariant's ML approach provides superior sensitivity critical for comprehensive variant detection in research and clinical settings.

VALIDATION OF BARBITOFF ET AL. (2022)

ORIGINAL PAPER FOUND	OUR STUDY FOUND
DeepVariant > GATK	✓ **DeepVariant > GATK**
Higher Sensitivity	✓ **6-14% More Variants**
GATK ⊆ DeepVariant	✓ **Zero GATK-only Variants**

↓

Despite limitations, trends match original paper → findings likely robust
Original paper also lacked SV analysis, runtime comparison, phasing.

Methodology

Successful reproduction factors:

- **Identical Software Versions**
  GATK 4.2.3.0 & DeepVariant 1.2.0
- **Benchmark Approach**
  Synthetic micro-benchmark on consistent datasets
- **Trend Alignment**
  Results match original study despite smaller scale

FINDINGS ARE ROBUST & REPRODUCIBLE

References

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Any Questions?

THANKS for your participation!!!

BINF-6310-Group-ART- Presentation
Reproducible Genomics